



Movie Challenge Lite

Looking for a movie trivia game? If yes, then Movie Challenge Lite is worth a shot
<http://dgit.in/PHkjLM>

No Facebook Phone

Mark Zuckerberg quelled all rumours about a potential smartphone from the social network by completely denying any such development

SoC, 512MB of RAM and 2GB of internal storage, while I/O ports like microSD slot, USB 2.0, VGA, HDMI, 3.5 mm audio and headphone jack are also supported. The VIA APC runs a slightly modified version of Android 2.3 Gingerbread with the ability to detect keyboard and mouse (Android doesn't natively support external keyboard or mouse). The form factor of the device is dubbed as Neo-ITX which is considerably smaller than mini-ITX boards. With a fully functional web browser, and the range of I/O devices supported by the APC, it truly is a tiny PC at an unbelievable price.

Just this month, we got our hands on a Chinese made Android 4.0 Mini PC which isn't any bigger than a USB pendrive, justifying the label of a Thumb PC to the fullest. Only a handful of connectivity ports are present on the slim



The VIA APC has a neo-ITX form factor

and tiny device – mini USB, microSD card slot, USB 2.0 port, and a mini HDMI port, not to mention built-in Wi-Fi. We tried connecting a USB keyboard and mouse through a hub and tried operating the Thumb PC and its Android 4.0 ICS interface. Obviously, it isn't the same as using a Windows-based PC, but with a little patience and the might of Google's Play store at your disposal, you can get by. Like any new product of an emerging category, the Android 4.0 Mini PC has its pros and cons (read the extensive review at <http://dgit.in/Py41Vv>) but for Rs. 5,000, it's a good second PC that you can take around wherever you go.

The Norwegian manufacturer FXI has started shipping Cotton Candy, which is just another example of the potential of these tiny computers. It measures just a tad over three inches and has a dual-core ARM processor. A micro USB port, mini HDMI port and a microSD slot make up



The pendrive-sized Android 4.0 mini PC

the physical I/O ports available on the device. It also has built-in Wi-Fi and Bluetooth radios for connecting a pair of wireless keyboard and mouse. You can either install Android 4.0 or Ubuntu on the Cotton Candy, which sells for a price of \$199 and can be used to make any TV a smart TV or extending your smartphone's screen environment.

While steady in number, the current state of Thumb PCs isn't really a mass market product simply because they aren't great examples of a PC for general use. No complaint about their form factors – they're fantastic – and that's the right direction to go, but on a hardware and software level it will continue to be a restrictive environment. Unless something changes fundamentally, and that change may well be in the offing.

What's in store

Obviously, the current wave of Thumb PCs is clearly piggy-backing on the phenomenal success of modern day smartphones and tablets, and we all have ARM's architecture (prevalent in over 90% of smartphones) to thank for. On current evidence, there is no way to tell whether Thumb PCs will make a significant dent in the market. But Intel, a giant from the PC

chip manufacturing sector, is expected to make huge inroads by offering a flavour of x86 computing on smartphones and tablets. It has already launched the Lava XOLO X900, an Android smartphone running on a 1.6 GHz Intel Atom Z2460 processor. According to one report, Intel's upcoming Medfield-based offerings will be better in power utilization and graphics performance compared to a competing ARM offering. All this points to a Thumb PC based on Intel's architecture in the near future.

While comparing hardware projections, don't forget the beauty of software. So far, Android and Linux has played an equally important role in re-igniting interest in Thumb PCs. But come October, Microsoft will reveal the latest version of its operating system, Windows 8, and with it support for products based on ARM architecture – which includes smartphones, tablets, and the present state of Thumb PCs. All this is just excellent news for the immediate future of Thumb PCs. Windows is the most widely-used operating system on the planet, and there's no ruling out the impact Microsoft's upcoming Windows 8 may have in boosting the widespread acceptability of Thumb PCs. While it may be struggling to woo developers on Windows Phone so far, it has a global army of developers for Windows x86 and x64, the traditional PC versions. Imagine the potential of a Thumb PC based on Intel's Medfield chips like the one on the Lava XOLO X900 (supporting x86 instruction set) running on Windows 8. That is a potent combination, and one can only hope that such a scenario does come to fruition.

All said, there's a lot of flux in the PC market. The PC as we knew it in the mid-90s was first conceived by IBM, and who's to say that PCs of the near future won't be radically different in design? In this age of digital convergence, we are habituated to expect accelerated performance on shrinking form factors. By all indications, and after tablets and hybrids, Thumb PCs resemble the latest step in the journey of personal computing. Linux will always be there, but for the explosive rise of Thumb PCs, a lot will ride on ARM, Intel and Microsoft's near-term future. **d**



FXI's Cotton Candy measures just 8 cm